

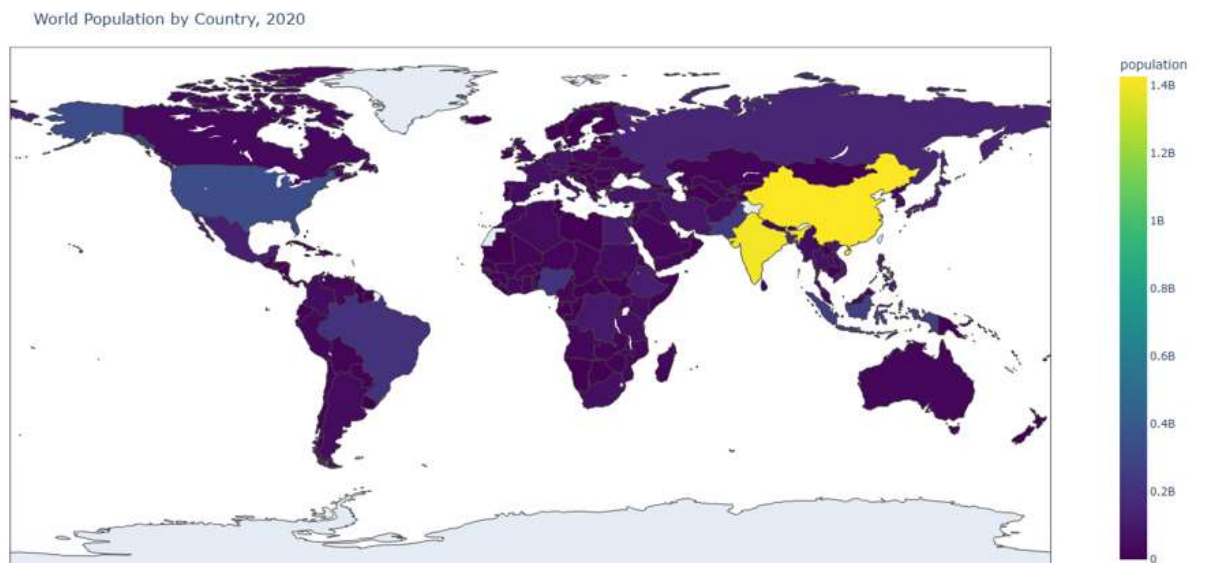
Population Size and Economic Prosperity: A 2020 Cross-Country Analysis

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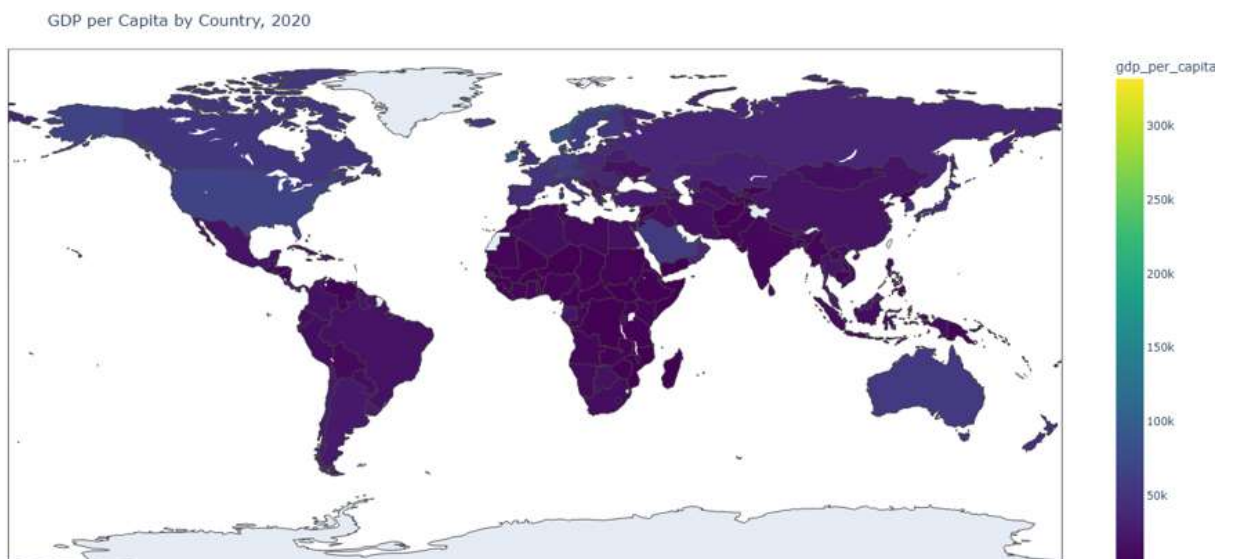
Date: 17.05.2026

Outputs

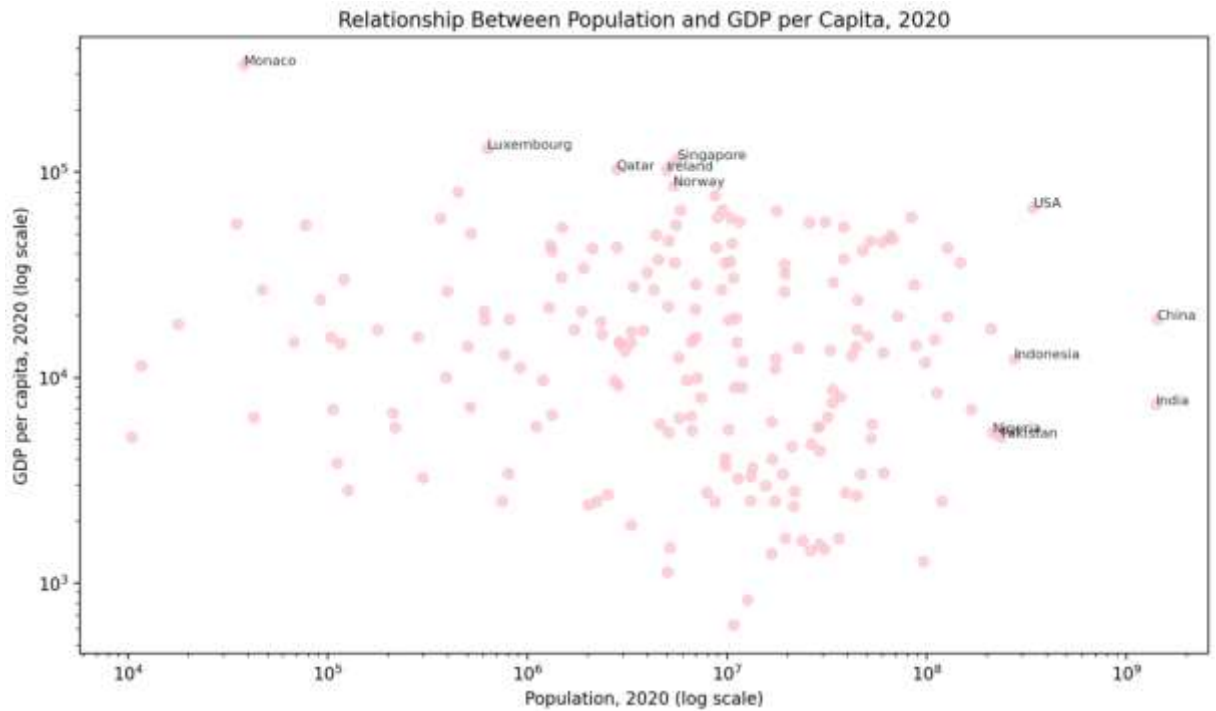
- [Population 2020 world map](#)



- [GDP per capita 2020 world map](#)



- [Population vs GDP per capita 2020 scatterplot](#)



Brief interpretation

The relationship between population and GDP per capita in 2020 appears **weakly negative**, but not strong. The raw Pearson correlation is **-0.044**, while the rank correlation is **-0.130**. On log-transformed values, the correlation is also weakly negative at **-0.125**.

This suggests that countries with larger populations are not necessarily richer or poorer on a per-person basis. Very large countries such as China, India, Indonesia, Nigeria, and Pakistan have moderate or lower GDP per capita compared with smaller high-income countries such as Monaco, Luxembourg, Qatar, Singapore, and Ireland. Overall, population size alone does **not** explain GDP per capita very well.

Reflection:

The process of interacting with the AI tools was surprisingly very easy, mainly that we were provided the correct prompt.

I changed the color of the Scatterplot from light blue to pink.

I also discovered how to generate an image (.png) from the maps rather than taking a screenshot, which was an additional learning experience.